SOC corrected Gaussian basis sets for electronic transport calculations in ANT.Gaussian.

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Introduction

This is very brief summary of the implementation discussed in Pakdel et al. [1]. There we showed that if a GTO basis set describes properly the band structure of a material in the absence of SOC, then adding SOC as a correction in a post self-consistent-field (SCF) step gives a good approximation to the actual bands when compared to reference methods which produce very high quality results such as `VASP` [2], `Wien2k` [3], `Quantum Espresso` [4] or `OpenMX` [5–8].

Details

The Dirac-Kohn-Sham Hamiltonian can be written, to lowest order, simply adding a standard atomic SOC contribution [1] to the SOC-free Kohn-Sham Hamiltonian:

$$\xi\left(r\right)\mathbf{L}\cdot\mathbf{S}.$$

When represented in an atomic orbital basis set, it becomes

$$\left[\xi_{\mathrm{ij}}\left\langle l_{\mathrm{i}};m_{\mathrm{i}};s|\mathbf{L}\cdot\mathbf{S}|l_{\mathrm{j}};m_{\mathrm{j}};s'\right\rangle\right], \tag{1}$$

where

$$\xi_{\mathrm{ij}}=\frac{e^{2}}{2m_{e}^{2}c^{2}}\int_{0}^{\infty}\frac{1}{r}\frac{\mathrm{d}V_{\mathrm{eff}}\left(r\right)}{\mathrm{d}r}R_{\mathrm{i}}\left(r\right)R_{\mathrm{j}}^{*}\left(r\right)r^{2}\mathrm{d}r. \tag{2}$$

In Eq. (2), $V_{\mathrm{eff}}\left(r\right)$ is the effective nuclear potential [1]. $R_{\mathrm{i}}\left(r\right)$ are the radial contracted gaussian-type orbitals (CGTOs). Only CGTOs on the same atom and of the same shell type ($L=1$, 2 or 3) contribute to the integral because SOC is an intra-atomic phenomenon [1]. However, for CGTO basis sets with pseudopotentials, a single multiplicative correction to ξ_{ij} is needed in order to account for the correct effective charge in $V_{\mathrm{eff}}\left(r\right)$ due to the lack of nodal structure near the nucleus in pseudopotentials [1]. Here, we make two minor modifications to improve the above implementation. We use the following Yukawa screening potential:

$$V_{\mathrm{eff}}\left(r\right)=\frac{\left(Z-1\right)\exp\left(-\frac{lnZ}{r_{c}}r\right)+1}{r} \tag{3}$$

where r_c is a cutoff, typically the size of an atomic radius ($\sim 2.5 - 3.0$ a.u.) and Z is the atomic number. The cutoff varies between shells, being larger for the inner-most shells and smaller for the outer-most ones. For simplicity, we will choose a common value that suits better the shell or shells dominating at the Fermi level (typically $L = 1$, 2 or 3), although multiplicative factors can be used to tune the strength of SOC for the other shells. For instance, for transition metals, the strength for $L = 1$ shells is usually 3-5 times that of the $L = 2$ shells [9] and this something that may not be caught by a common cutoff radius. .

Calibration of the SOC factors (2026)

The factors below replace those of the first version of this database. Those were fitted with a script that, when building $H(\mathbf{k})$ from the lower-triangular Fock matrices printed by `CRYSTAL`, counted every SOC matrix element above the diagonal twice (non-Hermitian $H(\mathbf{k})$); `ANT.Gaussian` adds the SOC term correctly, so those factors were too small for it. The present factors are obtained as follows:

(i) For each element, two `CRYSTAL23` [10] calculations of the bulk crystal with the basis set listed below and the Stuttgart small-core pseudopotential (`STUTSC` library): a scalar-relativistic one, and a self-consistent two-component one including the spin–orbit part of the same pseudopotential. Both share geometry, functional (PBE), $\mathbf{k}$ mesh and smearing.

(ii) The SOC term of Eqs. (1)–(3) is computed exactly as in `ANT.Gaussian` (same radial quadrature and units), added to the on-site block of the scalar-relativistic Hamiltonian, and $\mathrm{SOCFAC}_{P,D,F}$ are fitted by least squares to the two-component bands (same basis, so bands correspond one to one) between $E_F - 6$ and $E_F + 4$ eV, allowing a rigid shift. r_c is kept at 2 bohr; letting it free does not improve the fit.

The scalar part of `STUTSC` coincides term by term with the pseudopotentials listed below (checked for each element), so the factors apply to `ANT.Gaussian` runs with exactly these basis sets and pseudopotentials. For all-electron basis sets (Al, Cu) all factors are 1.

FCC Au

Basis set: the `Au_pob_TZVP_rev2` basis set plus an additional uncontracted s primitive (exponent 0.1) [11], with the Stuttgart small-core pseudopotential. Structure: a face-centred cubic (FCC) lattice with $a = 4.05$ Å. Reference: self-consistent two-component `CRYSTAL23` [10] calculation (PBE, $32 \times 32 \times 32$ Monkhorst–Pack mesh, Fermi smearing 0.001 Ha) with the same basis set and the spin–orbit part of the same pseudopotential (`STUTSC` library). The factors are fitted to its bands between $E_F - 6$ and $E_F + 4$ eV (path plus a $6 \times 6 \times 6$ mesh): RMS deviation 10.3 meV, maximum 48.9 meV, with r_c fixed at 2.0 bohr. The bands are insensitive to SOC_FAC_F, which is therefore taken from the projection of the on-site CRYSTAL SOC matrix onto the model.

On-site projection, for comparison: SOC_FAC_P = 231.2, SOC_FAC_D = 43.7, SOC_FAC_F = 12.6.

SOC factors (ANT keywords):

```
SOC_FAC_P = 201.3
SOC_FAC_D = 43.2
SOC_FAC_F = 12.6
RCUT      = 2.0
```

Basis set:

```
Au 0
S 3 1.0
30.0000000000 0.207492311080
27.0000000000 -0.332678933940
14.7468243310 0.383028179580
S 1 1.0
6.10568239000 1.000000000000
S 1 1.0
1.13068946000 1.000000000000
S 1 1.0
0.18001283000 1.000000000000
S 1 1.0
0.10000000000 1.000000000000
P 4 1.0
15.5000000000 0.150017118800
14.0000000000 -0.236098131830
6.42273682050 0.314588969480
1.65956016810 -0.572796704460
P 1 1.0
0.87155106000 1.000000000000
P 1 1.0
0.20887895000 1.000000000000
D 4 1.0
9.55240986560 0.040145559502
7.26988869370 -0.093690906606
1.77464967890 0.317462823170
0.79960541055 0.467951924830
D 1 1.0
0.55708019000 1.000000000000
D 1 1.0
0.28104926000 1.000000000000
```

```
F 1 1.0
0.72482000000 1.000000000000
```

Pseudopotential:

```
Au 0
AU-ECP 5 60
h potential
1
2 1.00000000 0.00000000
s-h potential
2
2 13.52321800 426.64186700
2 6.26438400 36.80066800
p-h potential
4
2 11.41386700 87.00209100
2 10.32921500 174.00437000
2 5.70742400 8.87061000
2 4.82816500 17.90243800
d-h potential
4
2 7.43096300 49.88365500
2 8.32199000 74.68454900
2 4.60964200 6.48622700
2 3.51150700 9.54682100
f-h potential
2
2 3.08463900 8.79164000
2 3.02474300 11.65845600
g-h potential
2
2 3.97844200 -5.23433700
2 4.01149100 -6.73814200
```

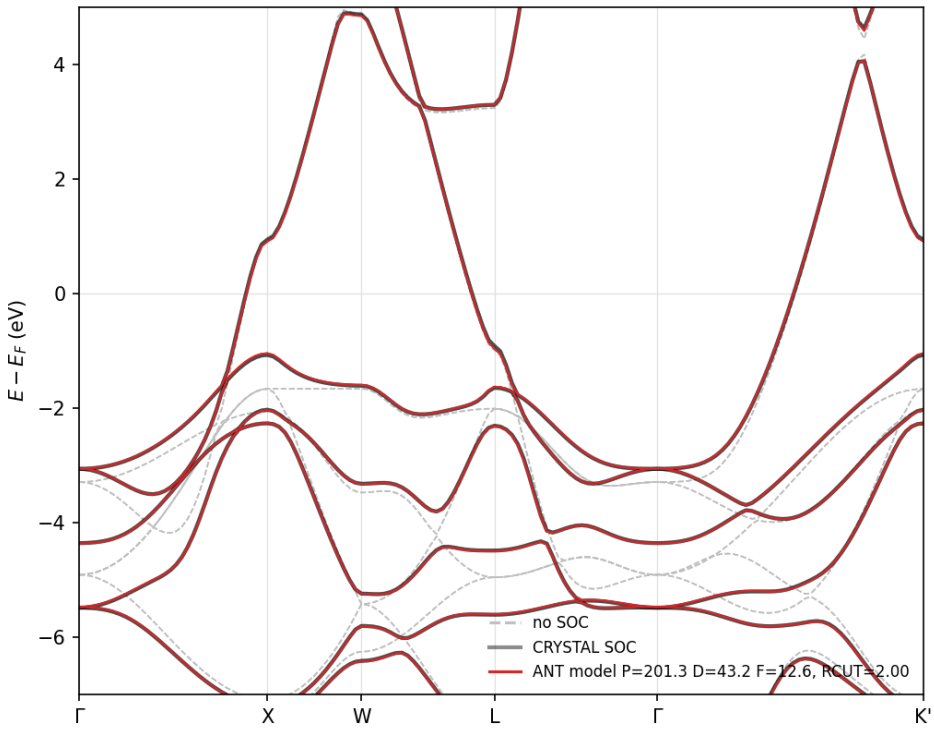


Figure 1. FCC Au: bands of the self-consistent two-component `CRYSTAL23` calculation (grey), of the no-SOC Hamiltonian plus the ANT on-site SOC with the factors above (red), and without SOC (dashed).

FCC Ag
Basis set: the `Ag_pob_DZVP_2018` basis set [12], with the Stuttgart small-core pseudopotential. Structure: an FCC lattice with $a = 4.075$ Å. Reference: self-consistent two-component `CRYSTAL23` [10] calculation (PBE, $32\times32\times32$ Monkhorst–Pack mesh, Fermi smearing 0.001 Ha) with the same basis set and the spin–orbit part of the same pseudopotential (`STUTSC` library). The factors are fitted to its bands between $E_F - 6$ and $E_F + 4$ eV (path plus a $6 \times 6 \times 6$ mesh): RMS deviation 3.4 meV, maximum 18.2 meV, with r_c fixed at 2.0 bohr. The bands are insensitive to SOC_FAC_F, which is therefore taken from the projection of the on-site CRYSTAL SOC matrix onto the model. The bands depend only weakly on SOC_FAC_P (RMS 3.4–4.1 meV for SOC_FAC_P between 0 and 85): the SOC splittings in the window come from the d shells. The value given is the least-squares optimum. On-site projection, for comparison: SOC_FAC_P = 84.7, SOC_FAC_D = 8.2, SOC_FAC_F = 0.1.

SOC factors (ANT keywords):
```
SOC_FAC_P = 49.1
SOC_FAC_D = 8.9
SOC_FAC_F = 0.1
RCUT      = 2.0
```

Basis set:
```
Ag 0
S 3 1.0
9.08844200000 -1.980891879700
7.54073100000 2.755451334700
2.79400500000 0.227154083810
S 1 1.0
1.23043317000 1.000000000000
S 1 1.0
0.18387114000 1.000000000000
P 4 1.0
4.45124000000 -0.993521037710
3.67526300000 1.050052523700
1.26106209050 0.647475325370
0.54212477498 0.256215507230
P 1 1.0
0.15011983000 1.000000000000
D 4 1.0
7.79566722920 -0.017042912377
2.89265102380 0.234461548030
1.24742732030 0.447658775330
0.49313817671 0.390649545600
D 1 1.0
0.15938987000 1.000000000000
F 1 1.0
1.39711000000 1.000000000000
```

Pseudopotential:
```
Ag 0
AG-ECP 4 28
g potential
```

```
1
2 1.0000000 0.0000000
s-g potential
2
2 12.5677140 255.0547710
2 6.9976620 36.9833930
p-g potential
4
2 11.3164960 60.7157050
2 10.9580630 121.4438890
2 7.1114000 10.1718660
2 6.7733190 20.4865640
d-g potential
4
2 8.9284370 29.5049380
2 11.1025670 44.0187360
2 5.5432120 5.3683330
2 3.9288350 7.4083750
f-g potential
2
2 11.0129130 -12.6234030
2 11.0198980 -16.7643270
```

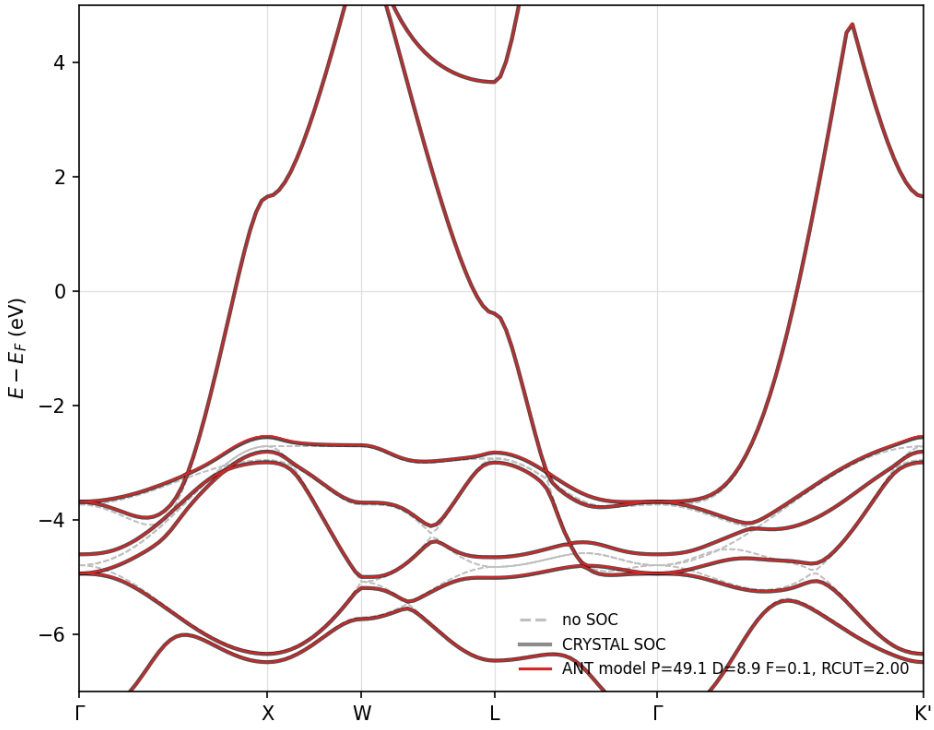


Figure 2. FCC Ag: bands of the self-consistent two-component `CRYSTAL23` calculation (grey), of the no-SOC Hamiltonian plus the ANT on-site SOC with the factors above (red), and without SOC (dashed).

FCC Pt

Basis set: the `Pt_pob_TZVP_rev2` basis set [11], with the Stuttgart small-core pseudopotential. Structure: an FCC lattice with $a = 3.91$ Å. Reference: self-consistent two-component `CRYSTAL23` [10] calculation (PBE, $32 \times 32 \times 32$ Monkhorst–Pack mesh, Fermi smearing 0.001 Ha) with the same basis set and the spin–orbit part of the same pseudopotential (`STUTSC` library). The factors are fitted to its bands between $E_F - 6$ and $E_F + 4$ eV (path plus a $6 \times 6 \times 6$ mesh): RMS deviation 5.0 meV, maximum 24.1 meV, with r_c fixed at 2.0 bohr. The bands are insensitive to SOC_FAC_F, which is therefore taken from the projection of the on-site CRYSTAL SOC matrix onto the model.

On-site projection, for comparison: SOC_FAC_P = 237.3, SOC_FAC_D = 42.6, SOC_FAC_F = 9.3.

SOC factors (ANT keywords):

```
SOC_FAC_P = 229.4
SOC_FAC_D = 41.6
SOC_FAC_F = 9.3
RCUT      = 2.0
```

Basis set:

```
Pt 0
S 3 1.0
  30.0000000000 0.271482639000
  27.0000000000 -0.422267587740
  14.4083185640 0.443615819950
S 1 1.0
  5.76260815000 1.000000000000
S 1 1.0
  1.06835871000 1.000000000000
S 1 1.0
  0.15368971000 1.000000000000
P 4 1.0
  15.5000000000 -0.156727186290
  14.0000000000 0.238534129890
  6.11612123390 -0.310413797330
  1.57155863850 0.564735250890
P 1 1.0
  0.77232013000 1.000000000000
P 1 1.0
  0.25285881000 1.000000000000
D 4 1.0
  8.32079376110 0.062945798646
  7.42072265200 -0.090271847072
  1.65704106390 0.168125264160
  0.73943569960 0.250454169700
D 1 1.0
  1.12524984000 1.000000000000
D 1 1.0
  0.27562365000 1.000000000000
F 1 1.0
  0.56813000000 1.000000000000
```

Pseudopotential:

```
Pt 0
PT-ECP 5 60
h potential
1
2 1.00000000 0.00000000
s-h potential
2
2 14.60450000 429.64608700
2 7.21828700 73.15688400
p-h potential
4
2 11.57716200 88.02291700
2 10.88384300 175.99819600
2 6.42440300 13.68227400
2 5.22419800 27.41465100
d-h potential
4
2 7.69961000 43.55785200
2 7.55080800 65.36910800
2 3.96116400 7.01859600
2 3.87277700 11.39173300
f-h potential
2
2 3.37986900 10.71022000
2 3.32625500 14.27812500
g-h potential
2
2 5.45202000 -11.65174900
2 5.41258500 -14.37552500
```

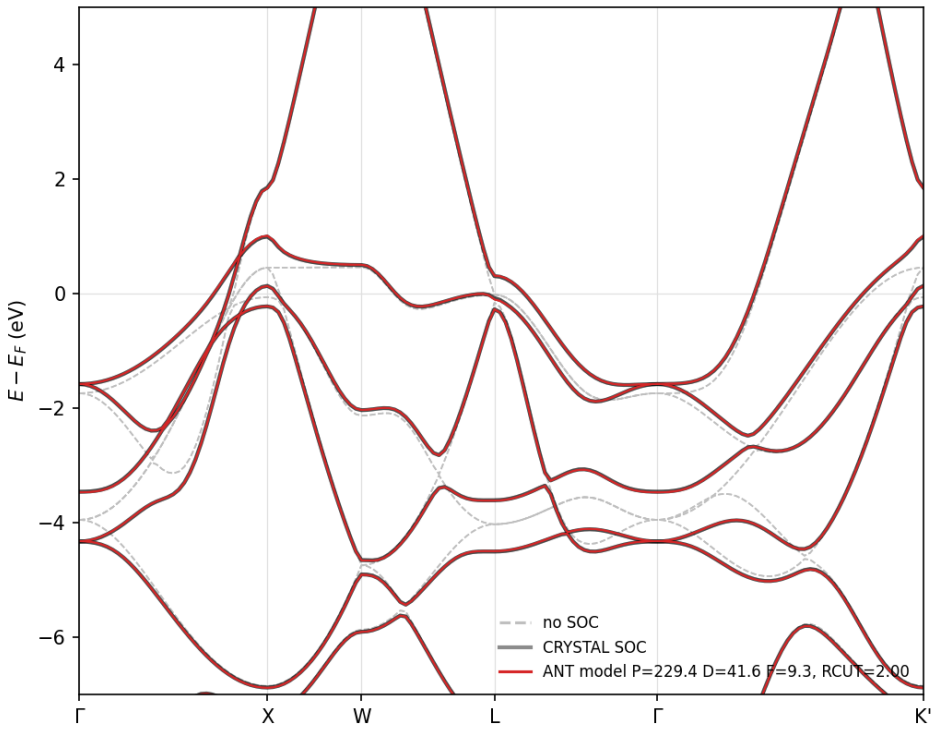


Figure 3. FCC Pt: bands of the self-consistent two-component `CRYSTAL23` calculation (grey), of the no-SOC Hamiltonian plus the ANT on-site SOC with the factors above (red), and without SOC (dashed).

BCC W

Basis set: the `W_pob_TZVP_rev2` basis set with additional diffuse s functions and re-optimized outer exponents [11], with the Stuttgart small-core pseudopotential. Structure: a body-centred cubic (BCC) lattice with $a = 3.16$ Å. Reference: self-consistent two-component `CRYSTAL23` [10] calculation (PBE, $32 \times 32 \times 32$ Monkhorst–Pack mesh, Fermi smearing 0.001 Ha) with the same basis set and the spin–orbit part of the same pseudopotential (`STUTSC` library). The factors are fitted to its bands between $E_F - 6$ and $E_F + 4$ eV (path plus a $6 \times 6 \times 6$ mesh): RMS deviation 5.8 meV, maximum 27.7 meV, with r_c fixed at 2.0 bohr. The bands are insensitive to SOC_FAC_F, which is therefore taken from the projection of the on-site CRYSTAL SOC matrix onto the model.

On-site projection, for comparison: SOC_FAC_P = 248.3, SOC_FAC_D = 55.1, SOC_FAC_F = 10.7.

SOC factors (ANT keywords):

```
SOC_FAC_P = 241.0
SOC_FAC_D = 55.4
SOC_FAC_F = 10.7
RCUT      = 2.0
```

Basis set:

```
W 0
S 3 1.0
  30.0000000000   0.322464834100
  27.0000000000   -0.466922572140
  13.0780456840   0.426995637760
S 1 1.0
  4.56489858000   1.000000000000
S 1 1.0
  0.92909758000   1.000000000000
S 1 1.0
  0.2029   1.000000000000
S 1 1.0
  0.117   1.000000000000
P 4 1.0
  17.3684   -0.0364
  12.4136   0.1092
  5.15862176580   -0.293999550200
  1.2804   0.5082
P 1 1.0
  0.6577   1.000000000000
P 1 1.0
  0.245   1.000000000000
D 4 1.0
  7.40647373150   0.086993963018
  5.90262686030   -0.176675400110
  1.29847567500   0.551456970300
  0.57153508541   0.953135965350
D 1 1.0
  0.4269   1.000000000000
D 1 1.0
```

```
  0.2466   1.000000000000
F 1 1.0
  0.3026   1.000000000000
```

Pseudopotential:

```
W 0
W-ECP 5 60
h potential
1
2 1.00000000 0.00000000
s-h potential
2
2 11.0637950 419.2275990
2 8.2176410 41.1913070
p-h potential
6
2 9.3381880 107.3481100
2 8.4304480 214.6995680
4 9.4900200 0.0254420
4 9.4899470 0.0518950
2 1.8829970 -0.1171840
2 1.9069720 0.2966890
d-h potential
6
2 6.2054330 58.8812790
2 6.1221570 98.6835560
4 6.2745560 0.0195370
4 6.2263750 0.0219560
2 1.9638750 -0.0885770
2 1.8882870 -0.2097260
f-h potential
2
2 2.3079530 6.2324720
2 2.2706090 8.3113450
g-h potential
2
2 3.5834910 -6.8029440
2 3.5625150 -8.4432320
```

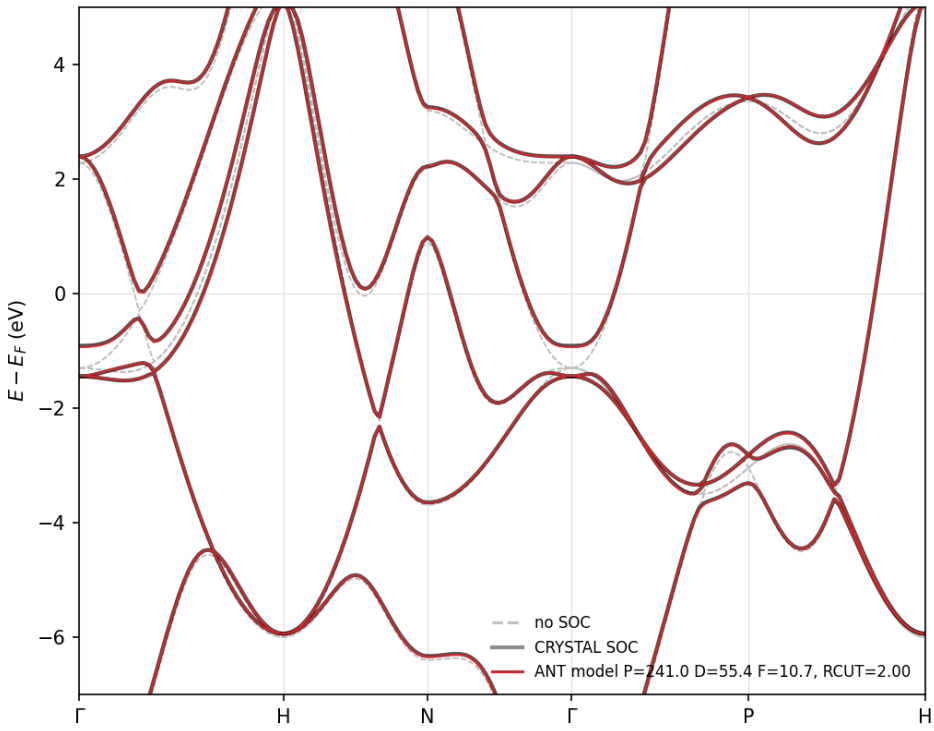


Figure 4. BCC W: bands of the self-consistent two-component `CRYSTAL23` calculation (grey), of the no-SOC Hamiltonian plus the ANT on-site SOC with the factors above (red), and without SOC (dashed).

FCC Pb

Basis set: the optimized basis set of Baranek and co-workers [13], with the Stuttgart small-core pseudopotential. Structure: an FCC lattice with $a = 4.93$ Å. Reference: self-consistent two-component `CRYSTAL23` [10] calculation (PBE, $32 \times 32 \times 32$ Monkhorst–Pack mesh, Fermi smearing 0.001 Ha) with the same basis set and the spin–orbit part of the same pseudopotential (`STUTSC` library). The factors are fitted to its bands between $E_F - 6$ and $E_F + 4$ eV (path plus a $6 \times 6 \times 6$ mesh): RMS deviation 9.9 meV, maximum 51.9 meV, with r_c fixed at 2.0 bohr. The bands are insensitive to SOC_FAC_F, which is fixed at 0. The pseudopotential is the complete Stuttgart small-core one (60 core electrons) including its g projector; the 2021 `CRYSTAL` input of this basis set omitted the two g terms. The projection gives a negative SOC_FAC_F, which ANT does not accept, hence 0. SOC_FAC_D is fixed by the bands even though the $5d$ shell lies below the window (RMS 18 meV for SOC_FAC_D = 0, 9.9 meV at the optimum). On-site projection, for comparison: SOC_FAC_P = 207.3, SOC_FAC_D = 38.1, SOC_FAC_F = −32.8.

SOC factors (ANT keywords):

```
SOC_FAC_P = 248.3
SOC_FAC_D = 66.5
SOC_FAC_F = 0.0
RCUT      = 2.0
```

Basis set:

```
Pb 0
S 6 1.0
  8.55351735988   3.88475299531
  7.35277444701   -5.11944720447
  2.95283552202   0.359342355194
  1.43101525276   2.0148141461
  0.659101026799   0.477751455641
  0.220157140772   -0.029791382408
S 1 1.0
  1.2543   1.
S 1 1.0
  0.4504   1.
S 1 1.0
  0.147   1.
P 6 1.0
  8.55351735988   3.13503035185
  7.35277444701   -4.65435922478
  2.95283552202   1.34852524939
  1.43101525276   3.23612718594
  0.659101026799   1.35196984634
  0.220157140772   0.0649906557872
P 1 1.0
  1.2879   1.
P 1 1.0
  0.5176   1.
P 1 1.0
  0.1325   1.
D 5 1.0
  11.1394028684   0.299749731795
```

```
  7.57317950313   -0.908307739883
  2.21243848295   3.44990840575
  1.0927431021   5.08865830004
  0.517878712738   3.46485836421
D 1 1.0
  0.2314   1.
F 1 1.0
  0.1354   1.0
```

Pseudopotential:

```
Pb 0
PB-ECP 5 60
h potential
1
2 1.00000000 0.00000000
s-h potential
2
2 12.2963030 281.2854990
2 8.6326340 62.5202170
p-h potential
4
2 10.2417900 72.2768970
2 8.9241760 144.5910830
2 6.5813420 4.7586930
2 6.2554030 9.9406230
d-h potential
4
2 7.7543360 35.8485070
2 7.7202810 53.7243420
2 4.9702640 10.1152560
2 4.5637890 14.8337310
f-h potential
2
2 3.8875120 12.2098920
2 3.8119630 16.1902910
g-h potential
2
2 5.6915770 -9.0966650
2 5.7155670 -11.5319960
```

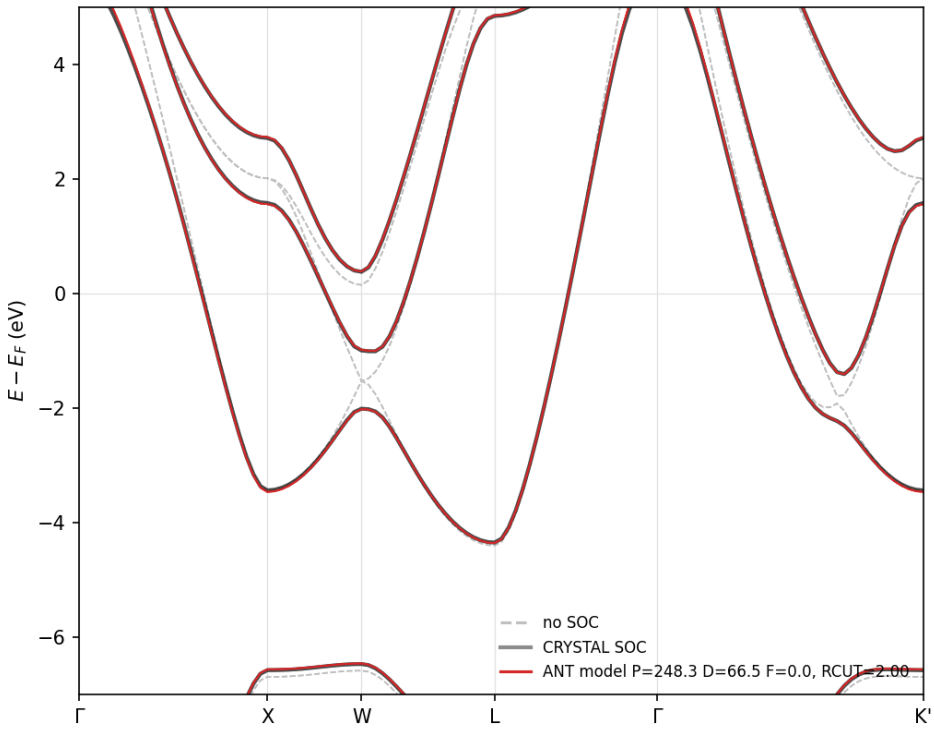


Figure 5. FCC Pb: bands of the self-consistent two-component `CRYSTAL23` calculation (grey), of the no-SOC Hamiltonian plus the ANT on-site SOC with the factors above (red), and without SOC (dashed).

Rhombohedral (A7) Bi

Basis set: the `Bi_pob_TZVP_rev2` basis set plus an additional uncontracted s primitive (exponent 0.1) [11], with the Stuttgart small-core pseudopotential. Structure: the rhombohedral A7 structure ($R\bar{3}m$) with the experimental room-temperature hexagonal parameters $a = 4.546$ Å, $c = 11.862$ Å and $u = 0.2339$. Reference: self-consistent two-component `CRYSTAL23` [10] calculation (PBE, $32 \times 32 \times 32$ Monkhorst–Pack mesh, Fermi smearing 0.001 Ha) with the same basis set and the spin–orbit part of the same pseudopotential (`STUTSC` library). The factors are fitted to its bands between $E_F - 6$ and $E_F + 4$ eV (path plus a $6 \times 6 \times 6$ mesh): RMS deviation 11.7 meV, maximum 41.1 meV, with r_c fixed at 2.0 bohr. The basis set has no f shells. SOC_FAC_P is well defined (229–233 for any SOC_FAC_D between 0 and 60), SOC_FAC_D only weakly (RMS 11.7–12.7 meV over that range). The on-site projection of SOC_FAC_P is dominated by the semicore $5p$ shell and is not representative of the valence $6p$ bands. On-site projection, for comparison: SOC_FAC_P = 114.9, SOC_FAC_D = 34.7.

SOC factors (ANT keywords):

```
SOC_FAC_P = 230.9
SOC_FAC_D = 24.1
SOC_FAC_F = 0.0
RCUT      = 2.0
```

Basis set:

```
Bi 0
S 4 1.0
  716.414353100   0.000312543071
  83.8060590470   0.001762476895
  21.1169628530   -0.219109834370
  15.4914481870   0.404112249310
S 2 1.0
  23.2398550290   -0.068255758685
  6.64742550000   0.978880464710
S 2 1.0
  1.96177440000   0.653863290000
  0.97252866000   0.346136700000
S 1 1.0
  0.20618896000   1.000000000000
S 1 1.0
  0.10000000000   1.000000000000
P 3 1.0
  15.2496446690   0.745603560000
  14.8461760530   -0.855786373380
  7.06368267840   0.401491595920
P 3 1.0
  2.58812556160   0.355427296330
  1.50202084990   0.639769918900
  0.76732724388   0.323327738390
P 1 1.0
  0.42288904000   1.000000000000
P 1 1.0
  0.17062087000   1.000000000000
D 6 1.0
```

```
  66.4044819480   0.000381028783
  13.8584269610   0.010746152442
  7.06545190000   -0.071947646845
  2.52521440350   0.261959749890
  1.34195850000   0.425947500000
  0.68340941000   0.336803256270
D 1 1.0
  0.90438000000   1.000000000000
D 1 1.0
  0.41457000000   1.000000000000
D 1 1.0
  0.15146817000   1.000000000000
```

Pseudopotential:

```
Bi 0
BI-ECP 5 60
h potential
1
2 1.00000000 0.00000000
s-h potential
2
2 13.0430900 283.2642270
2 8.2216820 62.4719590
p-h potential
4
2 10.4677770 72.0014990
2 9.1189010 144.0022770
2 6.7547910 5.0079450
2 6.2525920 9.9915500
d-h potential
4
2 8.0814740 36.3962590
2 7.8905950 54.5976640
2 4.9555560 9.9842940
2 4.7045590 14.9814850
f-h potential
2
2 4.2145460 13.7133830
2 4.1334000 18.1943080
g-h potential
2
2 6.2057090 -10.2474430
2 6.2277820 -12.9557100
```

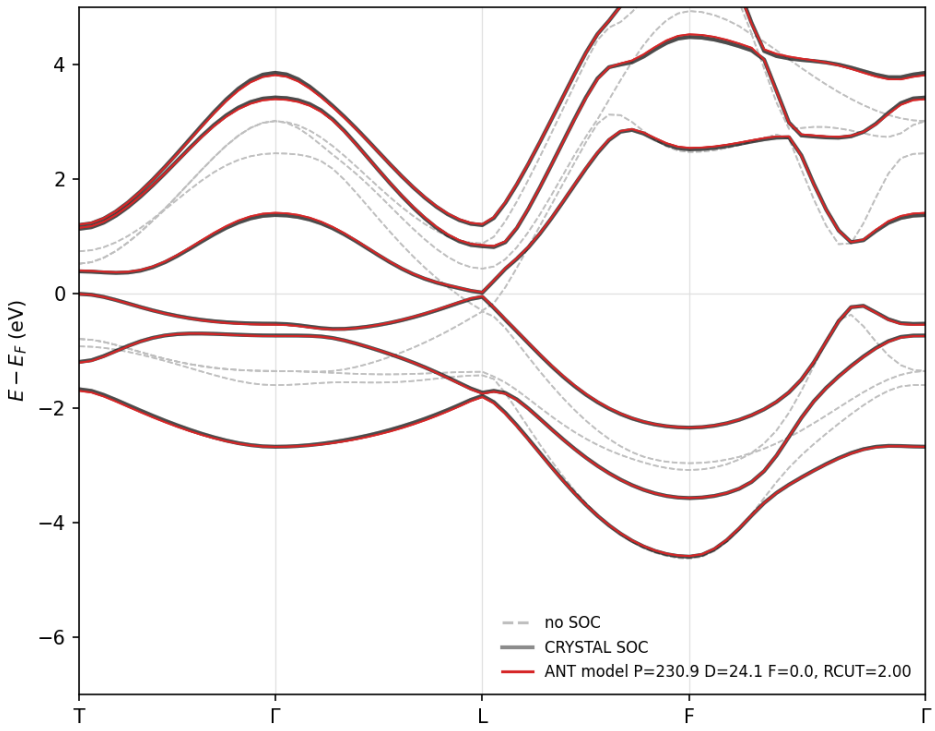


Figure 6. Rhombohedral (A7) Bi: bands of the self-consistent two-component `CRYSTAL23` calculation (grey), of the no-SOC Hamiltonian plus the ANT on-site SOC with the factors above (red), and without SOC (dashed).

Rhombohedral (A7) Sb

Basis set: the `Sb_pob_TZVP_2018` basis set [12], with the Stuttgart small-core pseudopotential. Structure: the rhombohedral A7 structure ($R\bar{3}m$) with the experimental room-temperature hexagonal parameters $a = 4.3007$ Å, $c = 11.222$ Å and $u = 0.2335$. Reference: self-consistent two-component `CRYSTAL23` [10] calculation (PBE, $32 \times 32 \times 32$ Monkhorst–Pack mesh, Fermi smearing 0.001 Ha) with the same basis set and the spin–orbit part of the same pseudopotential (`STUTSC` library). The factors are fitted to its bands between $E_F - 6$ and $E_F + 4$ eV (path plus a $6 \times 6 \times 6$ mesh): RMS deviation 5.8 meV, maximum 24.4 meV, with r_c fixed at 2.0 bohr. The basis set has no f shells. The bands prefer no SOC on the d shells: RMS 5.8 meV for SOC_FAC_D = 0, 6.3 meV at the on-site projection value 8.1 and 10.2 meV at 40 (SOC_FAC_P stays between 77 and 81). SOC_FAC_D ≈ 8 is an acceptable alternative if the semicore $4d$ splitting matters. On-site projection, for comparison: SOC_FAC_P = 38.9, SOC_FAC_D = 8.1.

SOC factors (ANT keywords):

```
SOC_FAC_P = 76.9
SOC_FAC_D = 0.0
SOC_FAC_F = 0.0
RCUT      = 2.0
```

Basis set:

```
Sb 0
S 4 1.0
  1612.41999330   0.000285403808
  238.844520970   0.001339377875
  23.9981188090   -0.049388154574
  15.1931242130   0.433922272540
S 2 1.0
  11.7364097330   0.921255199650
  6.52597747940   0.792352802260
S 1 1.0
  1.58348739000   1.000000000000
S 1 1.0
  0.18345132000   1.000000000000
P 3 1.0
  215.683933540   0.000260518232
  16.3744790880   0.073728000195
  9.72162833450   -0.272300281280
P 3 1.0
  2.79826431540   0.464726923740
  1.47110450330   0.503642420750
  0.75165385301   0.187066662940
P 1 1.0
  0.37211239000   1.000000000000
P 1 1.0
  0.12000001000   1.000000000000
D 6 1.0
  115.903122530   0.000531409151
  30.4742337200   0.005941113917
  18.2284182390   -0.010563706947
  4.32914566460   0.203481773410
```

```
  2.12948184960   0.427483789280
  0.99682636692   0.385395608090
D 1 1.0
  0.62366245000   1.000000000000
D 1 1.0
  0.31224235000   1.000000000000
D 1 1.0
  0.12000012000   1.000000000000
```

Pseudopotential:

```
Sb 0
SB-ECP 4 28
g potential
1
2 1.00000000 0.00000000
s-g potential
2
2 16.3308650 281.0715810
2 8.5565420 61.7166040
p-g potential
4
2 14.4703370 67.4573800
2 13.8161940 134.9335030
2 8.4249240 14.7163440
2 8.0927280 29.5185120
d-g potential
4
2 14.8863310 35.4478150
2 15.1463190 53.1434660
2 5.9082670 9.1792230
2 5.5943220 13.2402530
f-g potential
2
2 14.4449780 -15.3668010
2 14.4492950 -20.2961380
```

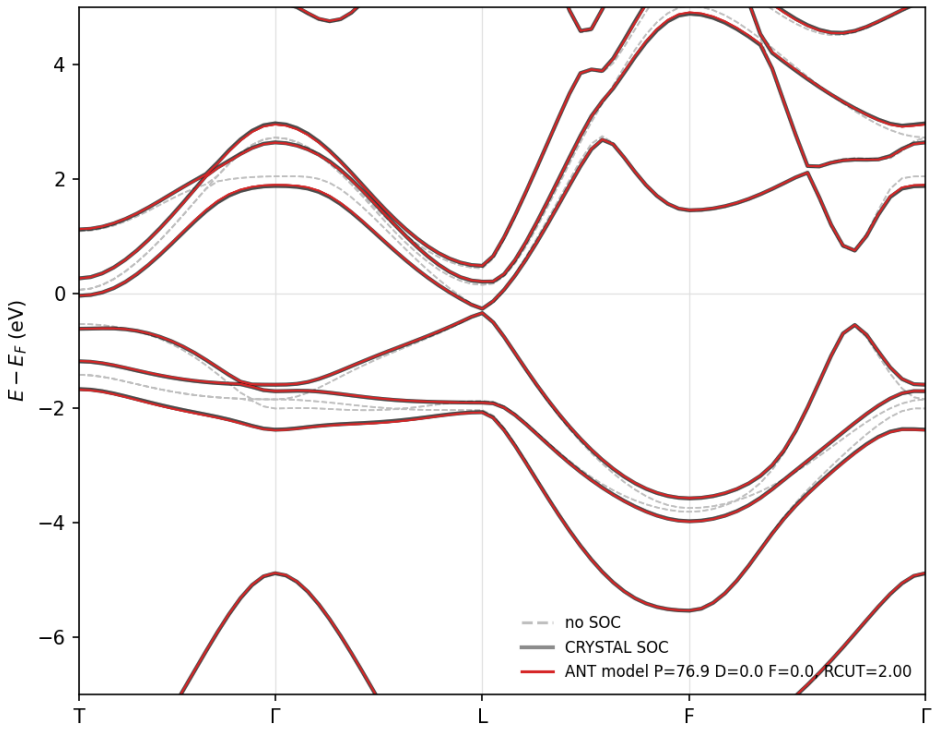


Figure 7. Rhombohedral (A7) Sb: bands of the self-consistent two-component `CRYSTAL23` calculation (grey), of the no-SOC Hamiltonian plus the ANT on-site SOC with the factors above (red), and without SOC (dashed).

FCC Al

Basis set: the all-electron `Al_pob_DZVP_rev2` basis set [14]. All-electron basis sets keep the nodal structure of the radial functions near the nucleus, which Eq. (2) assumes, so no multiplicative correction is applied (the 2021 fits against `OpenMX` also ended at factors of 1). Not recalibrated: `CRYSTAL23` includes SOC only through pseudopotentials with spin–orbit terms, so there is no same-basis reference for all-electron basis sets.

SOC factors (ANT keywords):

```
SOC_FAC_P = 1.0
SOC_FAC_D = 1.0
SOC_FAC_F = 0.0
RCUT      = 2.5
```

Basis set:

```
Al 0
S 5 1.0
5887.5727030 0.0013483347987
885.61225996 0.0100715768090
201.13604899 0.0451324540560
56.284974674 0.1146126804300
17.229551243 0.1015960894300
S 3 1.0
29.340249922 0.0693474542080
3.0439630420 -0.4252811767900
1.1285539518 -0.4144983221000
S 1 1.0
0.7834298000 1.0000000000000
S 1 1.0
0.1440019200 1.0000000000000
P 5 1.0
145.11918809 0.0063963373134
33.717894833 0.0441893599650
10.369863083 0.1558157599300
3.5135616036 0.2863528695100
1.1980050273 0.2292142324800
P 1 1.0
0.3200672400 1.0000000000000
P 1 1.0
0.1500336200 1.0000000000000
D 1 1.0
0.1530089300 1.0000000000000
```

FCC Cu

Basis set: the all-electron basis set of Ruiz *et al.* [15]. All-electron basis sets keep the nodal structure of the radial functions near the nucleus, which Eq. (2) assumes, so no multiplicative correction is applied (the 2021 fits against `OpenMX` also ended at factors of 1). Not recalibrated: `CRYSTAL23` includes SOC only through pseudopotentials with spin–orbit terms, so there is no same-basis reference for all-electron basis sets.

SOC factors (ANT keywords):

```
SOC_FAC_P = 1.0
SOC_FAC_D = 1.0
SOC_FAC_F = 0.0
RCUT      = 2.5
```

Basis set:

```
Cu 0
S 6 1.0
76441.483694 0.0014236548869
11477.790484 0.010910644019
2611.4773990 0.054139319495
737.33531101 0.18863206957
240.02819227 0.38341646126
82.721112667 0.29645681555
S 3 1.0
160.07447183 -0.11037499013
18.854087693 0.64586203973
7.7438526664 0.44511050812
S 2 1.0
13.678658833 -0.22684755916
2.2583597115 0.72397240174
S 1 1.0
.92525661579 1.0000000000
S 1 1.0
.41000000000 1.0000000000
S 1 1.0
.18000000000 1.0000000000
P 3 1.0
2530.0174404 0.0019142187743
600.09192696 0.015800163021
194.09448024 0.076268471159
P 3 1.0
73.686313161 0.23880092118
30.458500032 0.44974987602
13.130879763 0.39361952224
P 1 1.0
5.5263086949 1.0000000000
P 1 1.0
2.1487388219 1.0000000000
P 1 1.0
0.8145 1.0000000000
P 1 1.0
```

```
0.1327 1.0000000000
D 3 1.0
51.443304983 0.029319456167
14.404225180 0.15699236141
4.8477556014 0.37790966643
D 1 1.0
1.6194940205 1.0000000000
D 1 1.0
0.4664 1.0000000000
```

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The computational results contained in this work would also not have been possible without access to the high performance computing (HPC) facility at Unisa, and the supercomputing facility in the Department of Applied Physics at the University of Alicante.

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